

Distances Occurring Between One and n Times

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Abstract

Let $A \subset \mathbb{R}^2$ be a set of n points. We study distances determined by A whose multiplicity lies between 1 and n . We give a complete unconditional disproof of the literal statement that there must always be two such distances: an equilateral triangle determines only one distance. We also record the actual status of the stronger asymptotic question in the literature: known results on large distances do not yield the bound needed for a proof that $g(n) \rightarrow \infty$.

1 Statement of the problem

For a finite set $A \subset \mathbb{R}^2$, let

$$\Delta(A) := \{|x - y| : x, y \in A, x \neq y\}$$

be the set of distinct distances determined by A . For $d \in \Delta(A)$, write $m_A(d)$ for the number of unordered pairs $\{x, y\} \subset A$ with $|x - y| = d$.

We are interested in distances $d \in \Delta(A)$ satisfying

$$1 \leq m_A(d) \leq n.$$

Define

$$L(A) := \#\{d \in \Delta(A) : 1 \leq m_A(d) \leq n\}.$$

For each n , define the extremal quantity

$$g(n) := \min\{L(A) : A \subset \mathbb{R}^2, |A| = n\}.$$

The original questions are:

1. Must $g(n) \geq 2$?
2. Must $g(n) \rightarrow \infty$ as $n \rightarrow \infty$?

2 A small counterexample

Theorem 1. *The assertion $g(n) \geq 2$ is false in general.*

Proof. Take $n = 3$ and let A be the vertex set of an equilateral triangle. Then $\Delta(A)$ has exactly one element, namely the side length of the triangle, and that distance occurs exactly three times. Hence $L(A) = 1$, so $g(3) = 1 < 2$. $\square$

Thus the first question has a negative answer if interpreted literally for every n .

3 What the literature does and does not give

The argument in an earlier draft appealed to the following strong assertion: for each fixed k , the k largest distinct distances determined by a planar n -point set each occur at most n times. If this were available, then the asymptotic conclusions would follow immediately. However, this is not what the classical papers prove.

For clarity, define the following assertion.

For every fixed integer $k \geq 1$, there exists $N(k)$ such that for every $n \geq N(k)$ and every set $A \subset \mathbb{R}^2$ with $|A| = n$, if

$$\delta_1 > \delta_2 > \cdots > \delta_s$$

are the distinct distances determined by A in decreasing order, then

$$m_A(\delta_i) \leq n \quad (1 \leq i \leq k).$$

If this assertion is accepted, then for each fixed k the k largest distances all contribute to $L(A)$, and hence $g(n) \geq k$ for all sufficiently large n . In particular one would obtain both eventual validity of $g(n) \geq 2$ and the divergence $g(n) \rightarrow \infty$.

What is actually available online from standard references is weaker.

1. In *On large distances in planar sets* (Discrete Mathematics, 1987), Vesztergombi studied the multiplicity of the second-largest distance among n points in the plane. Later survey and expository sources summarizing this paper state that the second-largest distance can occur at most $\frac{3}{2}n$ times.
2. In later summaries of the large-distance literature, one finds the statement that the number of occurrences of the k -th largest distance in the plane is always smaller than $2kn$. This is again far weaker than the bound n required above.

Thus the statement needed by the earlier draft is not supplied by these references. In particular, the known bound $m_A(\delta_2) \leq \frac{3}{2}n$ does not even imply that the second-largest distance contributes to $L(A)$, because our threshold is exactly n .

Therefore the asymptotic claims are not merely missing a citation in the present note: they require genuinely stronger input than the classical bounds quoted above.

Remark 2. *There is therefore a sharp distinction between the two parts of the original problem. The first question has a complete unconditional negative answer, proved above. The stronger asymptotic claims require additional input not currently justified by the standard large-distance references discussed here.*

4 Conclusion

The present note proves the following unconditional statement.

Theorem 3. *Let $g(n)$ be defined as above.*

1. *The inequality $g(n) \geq 2$ is false for all n , since $g(3) = 1$.*
2. *No unconditional proof of the asymptotic statements is given in this note, and the standard large-distance bounds quoted in the literature do not suffice to prove them.*

References

- [1] K. Vesztergombi, *On large distances in planar sets*, Discrete Mathematics **67** (1987), 191–198.
- [2] P. Erdős, L. Lovász, and K. Vesztergombi, *On the graph of large distances*, Discrete & Computational Geometry **4** (1989), 541–549.
- [3] F. Morić and J. Pach, *Large simplices determined by finite point sets*, Beiträge zur Algebra und Geometrie **54** (2013), 481–487.